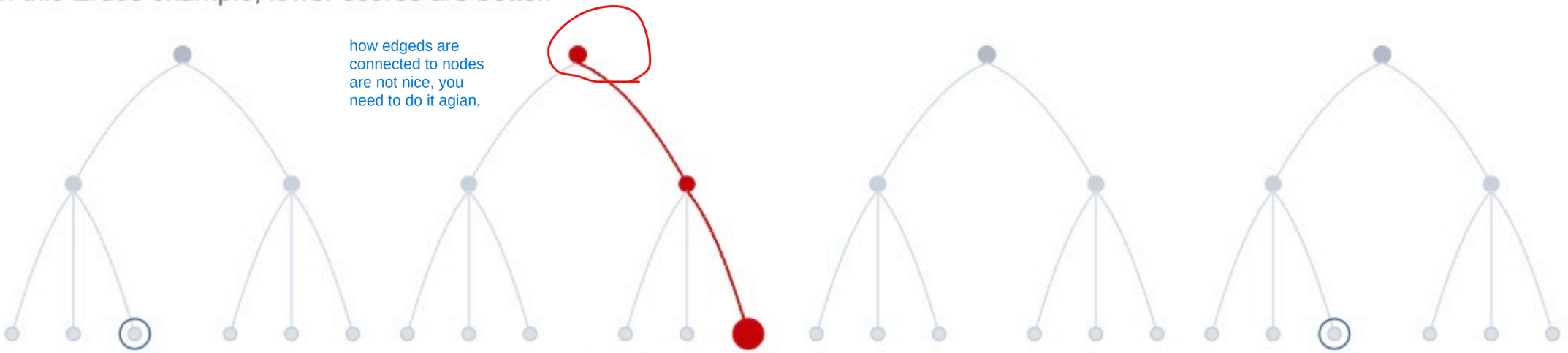
Technical Solution: Hierarchical Discovery Pipeline

A search for a better mathematical solution

01 / 06

In this Erdős example, lower scores are better.

how edgeds are connected to nodes are not nice, you need to do it agian,



Starting candidate (parent)

Score: 0.509541

Each dot is a saved candidate. Each link records where it came from.

A candidate is a mathematical construction produced by a program.

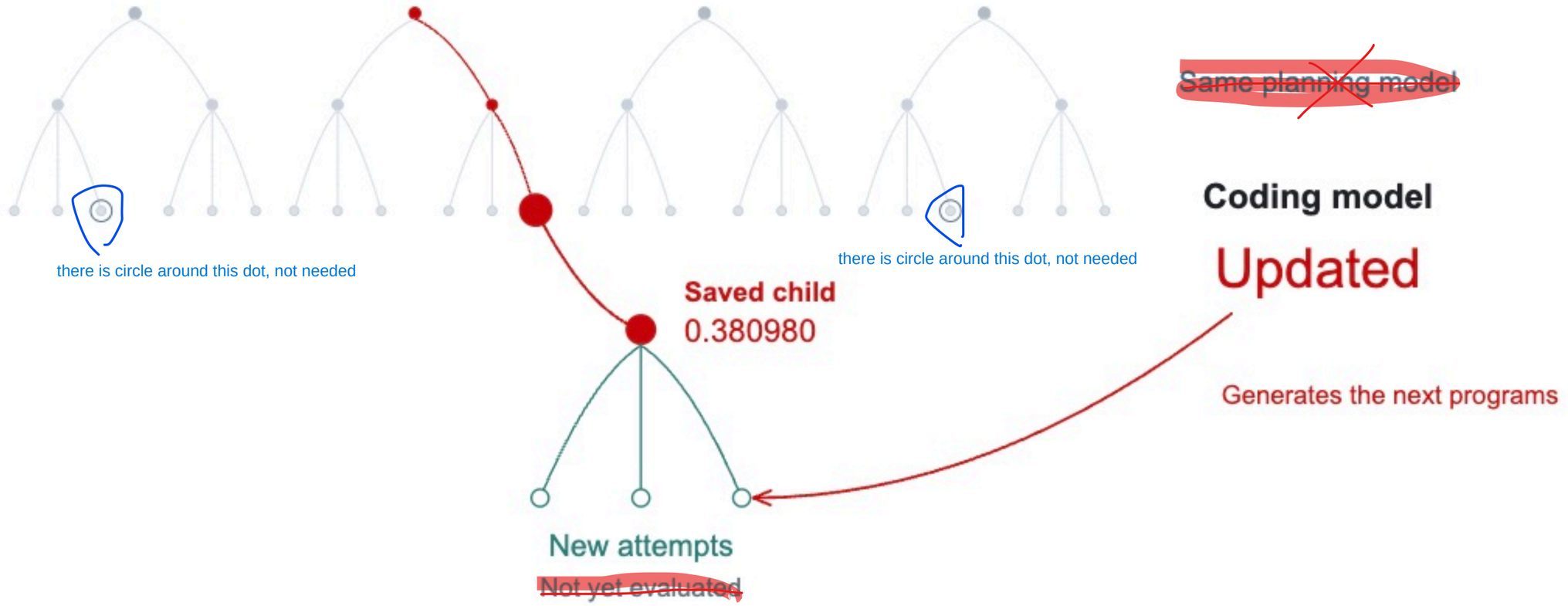
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Technical Solution: Hierarchical Discovery Pipeline

The search repeats with both updates

06 / 06

Saved candidates



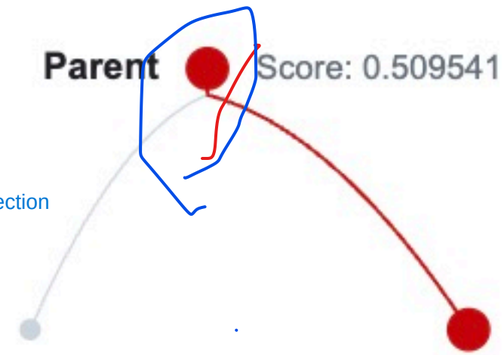
The next round uses saved candidates and the updated coding model. Every new candidate still needs evaluation.

Technical Solution: Hierarchical Discovery Pipeline

The system saves candidates and trains the coding model

05 / 06

Saved candidates (archive)



same issue of connection

New candidate (child)
0.380980

Retain top-r per strategy
then top-K per parent

Results from all parents

SPO² training

Coding model

Updated

Learns from evaluated programs

One update after the full batch

Historical reference fixed within the step

Planning model stays unchanged

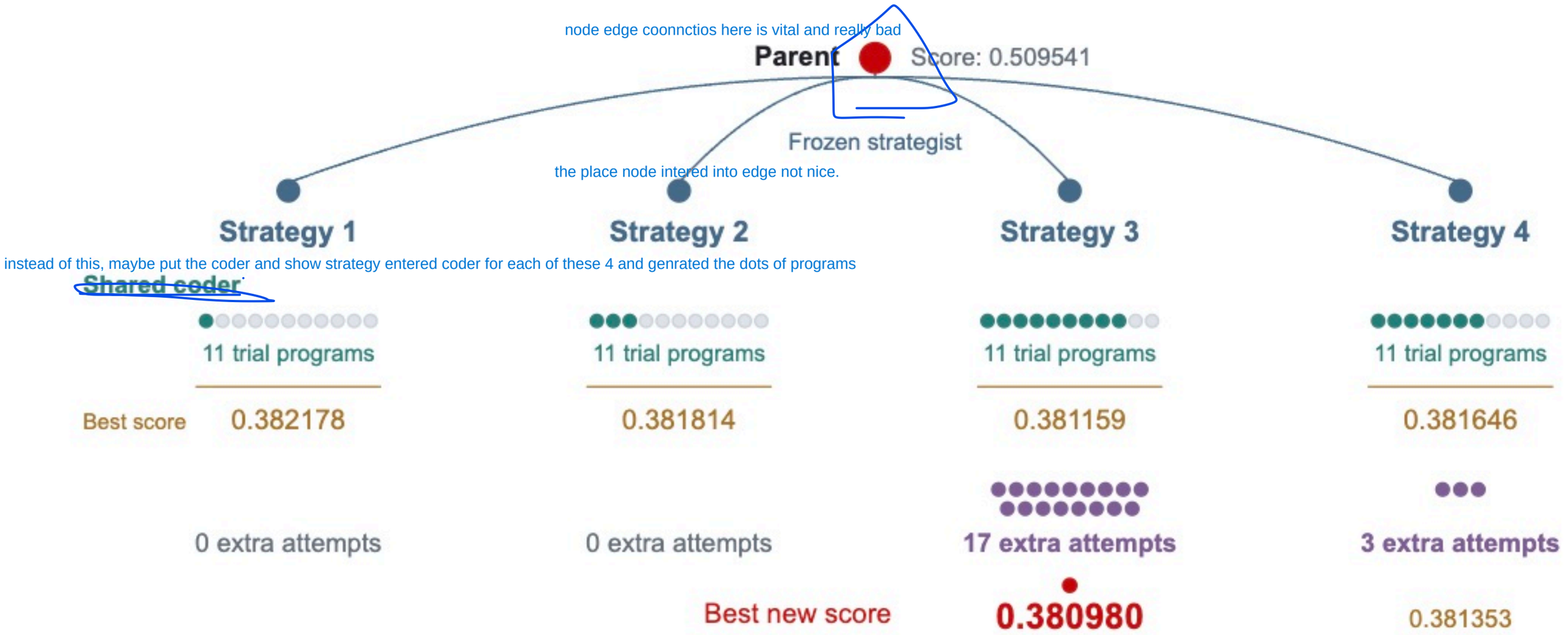
Two things change: useful candidates enter the archive, and evaluated programs train the coding model.

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Technical Solution: Hierarchical Discovery Pipeline

The bandit allocates more attempts using pilot results

node edge coonnctios here is vital and really bad



The bandit uses pilot results to assign ~~a fixed budget of 20~~ extra attempts, ~~called follow-ups~~

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Technical Solution: Hierarchical Discovery Pipeline

The coder turns each strategy into testable candidates



Each dot is one program

● Valid output ○ Invalid output

The evaluator runs the code, checks its output, and assigns a score.

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Technical Solution: Hierarchical Discovery Pipeline

Each strategy uses the parent and every earlier strategy



A strategy is a concrete plan. Each call receives the task, the same parent, and all strategies already generated for that parent.

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